

# Deploy LPWAN sensor networks to monitor and reduce energy consumption in mining operations.

Natural resources × Energy optimisation × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS164 v1 · refreshed 2026-08-18

| ATTRACTIVENESS                   | RIGHT TO WIN                         | COMPETITION                              | SERVICEABLE MARKET                           | HORIZON                                         | PORTFOLIO              |
|----------------------------------|--------------------------------------|------------------------------------------|----------------------------------------------|-------------------------------------------------|------------------------|
| <b>50</b><br>is the world moving | <b>46</b><br>can we play, can we win | <b>MEDIUM</b><br>0 named in the evidence | <b>€290m</b><br>modelled evidence · per year | <b>NEXT</b><br>pilots published no volume pr... | <b>L0</b><br>5 signals |

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

**Evidence gap (SC-13).** Orange has 2.0 published reference(s) in Natural resources, below the threshold of 5. A customer conversation here has no proof point behind it yet.

## The opportunity

Deploy LPWAN sensor networks to monitor and reduce energy consumption in mining operations. This opportunity targets mining operators seeking to decarbonize by first establishing reliable connectivity for energy monitoring and optimization. The need is urgent as the industry faces pressure to reduce emissions and improve operational efficiency.

## Market opportunity

| Method                                             | Total (TAM)                     | Serviceable (SAM)               | Obtainable (SOM)               | Confidence |
|----------------------------------------------------|---------------------------------|---------------------------------|--------------------------------|------------|
| Bottom-up: enterprises x adoption x contract value | <b>€388m</b><br>€78.6m – €1.7bn | <b>€290m</b><br>€58.7m – €1.3bn | <b>€4.4m</b><br>€881k – €18.8m | modelled   |
| Observed: tendered contract value, annualised      | €28.0m                          | €28.0m                          | €420k                          | observed   |

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

### Bottom-up: enterprises x adoption x contract value

| Factor                                              | Value                                                             | Basis      | Source                                        | As of          |
|-----------------------------------------------------|-------------------------------------------------------------------|------------|-----------------------------------------------|----------------|
| Enterprises in Natural resources (EU27_2020)        | 16,115 enterprises                                                | observed   | Eurostat · sbs_sc_ovw                         | 2024           |
| Enterprises adopting LPWAN (LoRaWAN, NB-IoT, LTE-M) | 42.6 % of enterprises (10+ employees)                             | proxy      | Eurostat · isoc_eb_iotn2 · E_IOT1             | 2021           |
| Annual value of one engagement                      | €320k per enterprise per year                                     | assumption | config/sizing.yaml · fallback_bands_eur       | size-2026-08-a |
| Size-weighted buyer base                            | 2,847 enterprise-equivalents at large-enterprise engagement value | assumption | config/sizing.yaml · size_class_value_weights | size-2026-08-a |
| Assumed obtainable share of the serviceable market  | 1.5 % of SAM                                                      | assumption | config/sizing.yaml · obtainable_share         | size-2026-08-a |

**Read this before quoting the number.** Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E\_IOT1 is used as a proxy for LPWAN (LoRaWAN, NB-IoT, LTE-M): IoT adoption as the connectivity demand proxy. Part of this vertical is not covered separately by the enterprise ICT survey, so the all-activities rate was used for those slices. Contract value is a configured assumption, not observed. Only 1 matching tender notices, below the 6 needed for a robust median, so the configured band for this business domain is used. It is an assumption owned by innovation-radar-curator, not evidence. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 2 declared geographies are outside the European reference data (AU, SO); the estimate covers the European footprint only. Sector adoption is read from Eurostat's all-activities aggregate for part of this vertical, which the enterprise ICT survey does not cover separately: PROXY — ICT survey excludes mining; PROXY — metal ores; PROXY — mining support; PROXY — oil and gas extraction; PROXY — other mining

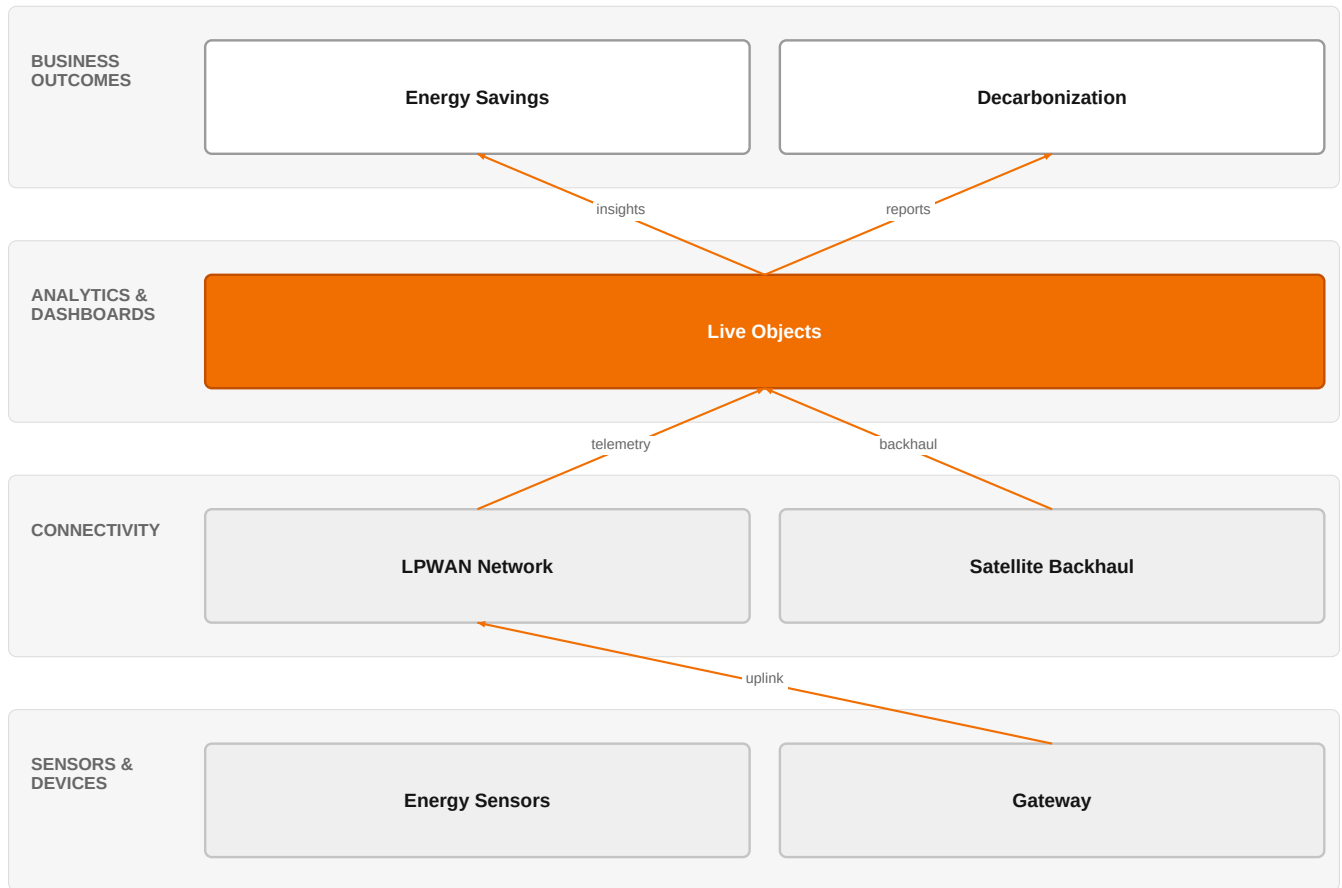
### Observed: tendered contract value, annualised

| Factor                                             | Value           | Basis      | Source                                | As of                  |
|----------------------------------------------------|-----------------|------------|---------------------------------------|------------------------|
| Tendered contract value in matching CPV categories | €28.0m per year | observed   | TED (Tenders Electronic Daily) · ted  | 2026-05-19..2026-07-22 |
| Assumed obtainable share of the serviceable market | 1.5 % of SAM    | assumption | config/sizing.yaml · obtainable_share | size-2026-08-a         |

**Read this before quoting the number.** Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 7 matching notices, matched on vertical via the CPV crosswalk — \$4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 2 declared geographies are outside the European reference data (AU, SO); the estimate covers the European footprint only.

## The solution, in one picture

### Energy Monitoring via LPWAN



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's Live Objects platform integrates with third-party sensors and connectivity to deliver energy savings and decarbonization outcomes.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

## What has changed

The mining industry is under increasing pressure to decarbonize, and the first step is establishing reliable connectivity to support energy monitoring and optimization. Advances in LPWAN technologies (LoRaWAN, NB-IoT, LTE-M) and satellite connectivity are enabling cloud-based mining operations, making it feasible to deploy sensor networks in remote and hazardous environments. Research highlights the role of AI, IoT, and advanced sensors in reshaping mining safety and efficiency, and adaptive federated learning is being proposed to optimize energy consumption in industrial wireless sensor networks.

Sources: SIG-5CEFDDFAA80A mining-technology.com 2026-08-14; SIG-CB178ABDC646 Discovery Alert 2026-08-11; SIG-97250BAF9D1E openalex.org 2025-10-07; SIG-395E0DE089C6 openalex.org 2025-11-12

## Who buys, and why now

The primary buyer is the Chief Sustainability Officer or Head of Operations at a mining company, who feels the pain of rising energy costs and regulatory pressure to reduce carbon emissions. The trigger is often a new emissions reduction target or a failed energy audit that reveals significant inefficiencies. They need to demonstrate progress on decarbonization, and they recognize that without reliable connectivity, they cannot even measure their energy consumption accurately.

## Why it is hot now — every claim bound to a dated source

- Mining decarbonisation efforts begin with establishing reliable connectivity to support energy monitoring and optimization. [SIG-5CEFDFAA80A]

## What Orange would deliver, and why it can win

### What Orange would deliver

Orange would assemble a solution using its IoT experts, Digital experts, and Orange Group researchers, leveraging the Live Objects platform for device management and data integration. The engagement would start with a connectivity assessment, then deploy LPWAN sensors (via partners) across mining sites, connect them through Live Objects, and provide dashboards for energy monitoring. Orange's Defense and security experts would ensure the solution meets sovereign and security requirements.

### Why Orange can win

Orange can win by combining its expertise in IoT, digital, and security with the Live Objects platform, which provides a scalable and secure foundation for sensor data. The involvement of Orange Group researchers brings innovation, and the defense and security experts ensure trust and sovereignty, which are critical for mining operations. However, the asset list is thin; Orange does not supply the sensors or connectivity infrastructure, so the win must be built on integration and platform strength.

### Named assets behind this — each individually inspectable (LK-08)

| Asset                        | Type            | Link | Why it is linked                                           | Curator     |
|------------------------------|-----------------|------|------------------------------------------------------------|-------------|
| Live Objects                 | offer           | L0   | offer addresses the use case AND provides the technology   | unconfirmed |
| IoT experts                  | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Digital experts              | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Defense and security experts | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |
| Orange Group researchers     | capability pool | SUP  | capability pool staffs this domain, vertical or technology | unconfirmed |

## Competitive landscape

**MEDIUM** competitive intensity (score 4.6; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

| Competitor        | Category                   | Position here                                                                                  | Basis      | Evidence |
|-------------------|----------------------------|------------------------------------------------------------------------------------------------|------------|----------|
| Telia Company     | Telecom operator peer      | sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries | structural | —        |
| Vodafone Business | Telecom operator peer      | sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries | structural | —        |
| Actility          | Industrial / OT specialist | sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries | structural | —        |
| Honeywell         | Industrial / OT specialist | active in Natural resources; competes in OX: Smart Industries                                  | structural | —        |
| Siemens           | Industrial / OT specialist | active in Natural resources; competes in OX: Smart Industries                                  | structural | —        |
| Marlink           | Satellite operator         | active in Natural resources; competes in Connectivity Solutions                                | structural | —        |
| Nokia             | Network equipment vendor   | active in Natural resources; competes in Connectivity Solutions, OX: Smart Industries          | structural | —        |
| SpaceX Starlink   | Satellite operator         | active in Natural resources; competes in Connectivity Solutions                                | structural | —        |

**evidenced** means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

## Qualifying questions for the first meeting

- 1 What is your current approach to monitoring energy consumption across your mining sites?
- 2 Have you already deployed any IoT sensors or connectivity solutions? If so, what are the main challenges?
- 3 What are your decarbonization targets and what timeline are you working towards?
- 4 How do you currently handle connectivity in remote or underground areas of your operations?
- 5 Who within your organization is responsible for energy management and what tools do they use today?

## Objections you will meet

| They will say                                              | You can answer                                                                                                                                                                         |
|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| We already have a solution from a major industrial vendor. | That's true, and they offer strong hardware. But our platform, Live Objects, can integrate with existing sensors and provide a unified view, potentially reducing complexity and cost. |
| LPWAN connectivity is not reliable in our remote sites.    | Reliability is a valid concern. We can combine LPWAN with satellite backhaul, as seen in the industry, to ensure coverage. Our experts can design a hybrid approach.                   |
| We don't have the budget for this.                         | Budget is always a constraint. However, the energy savings from optimized operations often offset the investment. We can start with a pilot to demonstrate value.                      |

## Risks and unknowns

The main risk is that mining customers may prefer a single-vendor solution from an industrial specialist like Siemens or Honeywell, who can provide sensors, connectivity, and analytics in one package. Another risk is that the business case may not be compelling if energy savings are not proven. Unknowns include the actual energy savings achievable and the reliability of LPWAN in harsh mining environments.

## Next action, by role (FR-17)

| Role                   | Do this next                                                                                                                                                                                                                                                       |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presales / Proposal    | Prepare a bid brief that differentiates Orange's approach by leveraging Live Objects and our IoT experts to address the specific challenges of energy monitoring in remote mining environments, and identify what additional evidence is needed to support claims. |
| Sales                  | In a conversation with a mining operations director, say: 'We can help you establish reliable connectivity for energy monitoring across your sites, using our Live Objects platform to manage LPWAN sensors and data.'                                             |
| Strategist / Innovator | Commission a deep-dive brief on how Orange's Live Objects platform and IoT expert pool can underpin LPWAN-based energy monitoring for mining operations, focusing on the connectivity reliability gap.                                                             |

## Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

| Signal           | Date       | Publisher             | T | Title and link                                                                                     |
|------------------|------------|-----------------------|---|----------------------------------------------------------------------------------------------------|
| SIG-CEB15C487D86 | 2026-01-12 | openalex.org          | 1 | Mobile Network Softwarization: Technological Foundations and Impact on Improving Network Energy... |
| SIG-395E0DE089C6 | 2025-11-12 | openalex.org          | 1 | Adaptive Federated Learning for Industrial Wireless Energy Optimization                            |
| SIG-97250BAF9D1E | 2025-10-07 | openalex.org          | 1 | Investigating advanced technologies to enhance worker safety in hazardous mining environments: ... |
| SIG-5CEFDDFAA80A | 2026-08-14 | mining-technology.com | 2 | Why mining decarbonisation challenge starts with reliable connectivity                             |
| SIG-CB178ABDC646 | 2026-08-11 | Discovery Alert       | 2 | Satellite Connectivity Powering Cloud-Based Mining Operations - Discovery Alert                    |

## How this brief was made

**Computed, not written.** Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

**Written, then checked.** The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive\_landscape (no claim survived evidence binding)

| What                        | Version                     | When                      |
|-----------------------------|-----------------------------|---------------------------|
| Scores — weight set         | w-2026-08-a                 | 2026-08-18T19:08:26+00:00 |
| Market sizing — assumptions | size-2026-08-a              | 2026-08-18T21:06:14+00:00 |
| Competitor register         | competitors-2026-08-a       | 2026-08-18T21:06:18+00:00 |
| Narrative — prompt / model  | describe-v1 / deepseek-chat | 2026-08-19T04:57:36+00:00 |
| Pipeline                    | 0.1.0                       | 2026-08-18                |

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.